



2020

TRIAL – YEAR 12

HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General

Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations

Total marks:

100

Section I – 10 marks (pages 2-5)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6-34)

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

Section I - 10 marks**Allow 15 minutes for this section**

1. Which expression is equal to $\int \tan^2 x \, dx$?

(A) $\frac{\tan^3 x}{3} + C$

(B) $\tan x - x + C$

(C) $\tan x + x + C$

(D) $\sec^2 x + C$

2. $\frac{d}{dx} \log_e \frac{4x^2 - 9}{2x - 3}$ is equal to which of the following?

(A) $\frac{6}{2x - 3}$

(B) $\frac{2}{2x + 3}$

(C) $\frac{6(2x + 3)}{(2x - 3)^2}$

(D) $\frac{6(4x + 1)}{(2x - 3)^2}$

3. Which of the following could be a primitive for $f'(x) = \frac{x}{e^{x^2 - 8}}$?

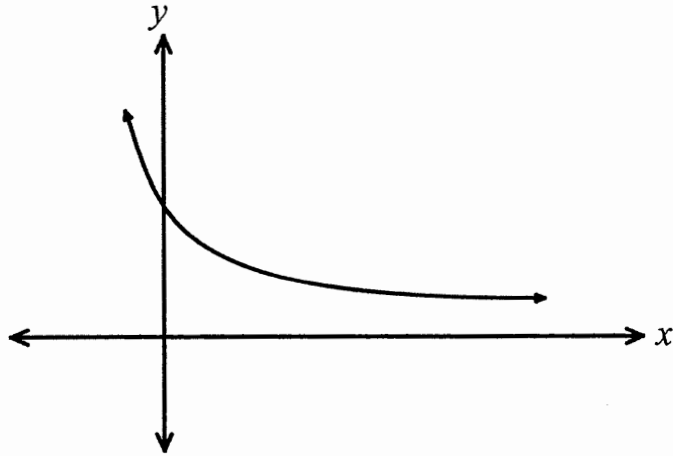
(A) $-\frac{1}{2}(e^{x^2 - 8}) + 8$

(B) $\frac{1}{2} \ln(e^{x^2 - 8}) + 8$

(C) $\ln(e^{8 - x^2}) - 8$

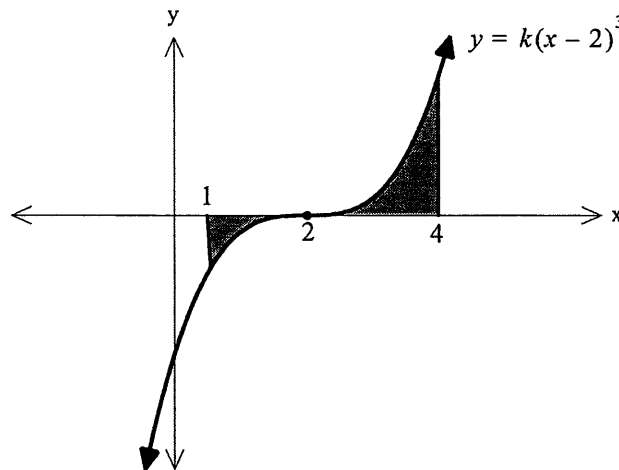
(D) $-\frac{1}{2}(e^{8 - x^2}) - 8$

4. For the curve shown, which inequalities are correct?



- (A) $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} > 0$
- (B) $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} < 0$
- (C) $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} < 0$
- (D) $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} > 0$
5. Results for a test are given as z-scores. In this test Angela gained a z- score of 3. The test has a mean of 55 and standard deviation of 6. What was Angela's actual mark in this test?
- (A) 58
- (B) 73
- (C) 64
- (D) 67

6. The graph with the equation $y = k(x - 2)^3$ is shown below, for some positive constant k .



If the area of the shaded region is 34, what is the value of k ?

- (A) $\frac{136}{15}$
- (B) 8
- (C) 4
- (D) $\frac{34}{9}$
7. The time, T , in seconds that divers can hold their breath is normally distributed with $\mu = 120$ and $Var(T) = 400$. In what range of time length would you expect to find the middle 95%?
- (A) $100 \leq x \leq 140$
- (B) $80 \leq x \leq 160$
- (C) $60 \leq x \leq 180$
- (D) $40 \leq x \leq 200$

8. The exact value of $I = \int_1^2 \frac{\ln x}{x} dx = \frac{1}{2}(\ln 2)^2$. The approximation of I using the Trapezoidal Rule with 2 function values is
- (A) smaller by 28%
 - (B) larger by 28%
 - (C) smaller by 72%
 - (D) larger by 72%
9. Given a function $f(x) = \frac{x}{x^2 - 5}$
- Which of the following statements is true?
- (A) $f(x)$ is even and one-to-one.
 - (B) $f(x)$ is even and many-to-one.
 - (C) $f(x)$ is odd and one-to-one.
 - (D) $f(x)$ is odd and many-to-one.
10. The amount M of certain medicine present in the blood after t hours is given by $M = 9t^2 - t^3$ for $0 \leq t \leq 9$.
- When is the amount of medicine in the blood increasing most rapidly?
- (A) $t = 0$
 - (B) $t = 9$
 - (C) $t = 6$
 - (D) $t = 3$

END OF SECTION I

Section II- Extended Response

Attempt Questions 11-16.

Allow about 2 hours and 45 minutes for this section.

Question 11(15 Marks)

a) Differentiate the following

(i) $y = (4x - 5)(4x + 5)$

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(ii) $y = \sin^2 x$

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b) In an arithmetic series, the third term is 5 and the tenth term is 26. Find the sum of the first 14 terms. 2

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Question 11 continued on the next page

c) Evaluate

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$$\int_1^4 5(9x - 4)^4 dx$$

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d) Solve the following equation for x .

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$$e^{2x} + 3e^x - 10 = 0.$$

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Question 11 continued on the next page

- e) (i) Show that $\frac{d}{dx}(\sec^2 x) = 2 \tan x \sec^2 x$. 2

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- (ii) Hence find $\int \tan x \sec^2 x \, dx$. 1

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Question 11 continued on the next page

f) Given a function $f(x) = \begin{cases} 6x - 6x^2 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$

(i) Show that $f(x)$ represents probability density function.

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(ii) Find the mode of the probability density function $f(x)$.

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End of Question 11

Question 12 (13 Marks)

- a) Find the value(s) of b such that $y = 2x + b$ is a tangent to the parabola 2

$$y = 2x^2 + 6x - 5.$$

[illegible]

Question 12 continued on the next page

- b) Angela guesses three questions in her multiple choice test, which has four options per question. Find the probability that Angela gets:

(i) Only one correct answer.

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(ii) At least one correct answer.

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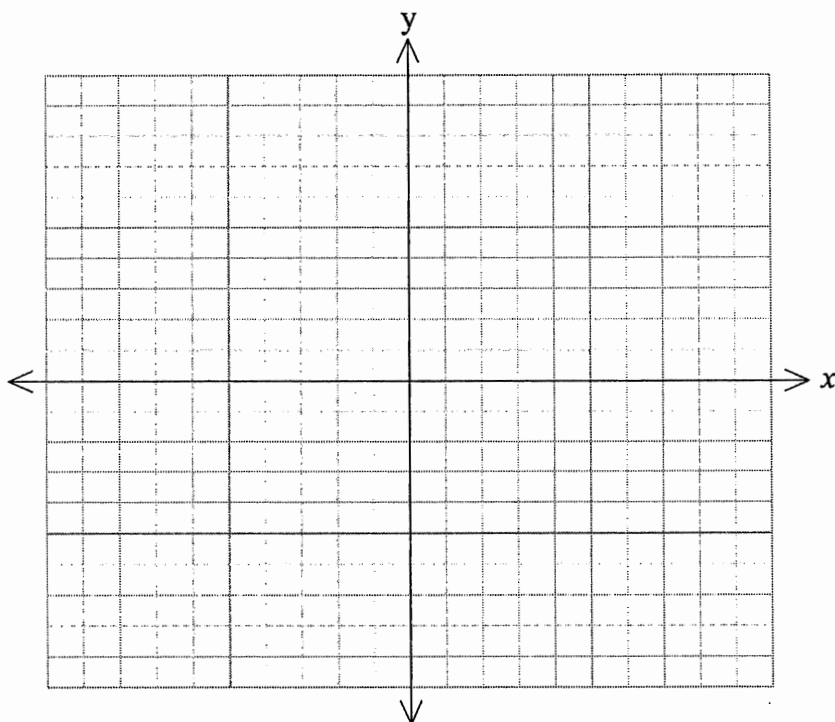
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Question 12 continued on the next page

c)

- (i) Sketch the hyperbola by shifting $y = \frac{1}{x-1}$ horizontally 3 units to the right 2
and 1 unit down.



- (ii) State the equation of the shifted hyperbola, then find all the intercepts 2
of the shifted hyperbola with the axes and mark them on your graph in
part (i).

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Question 12 continued on the next page

d) Consider the piece -wise defined function.

$$f(x) = \begin{cases} x^2 - 1 & x \leq 1 \\ 4 - x^2 & x > 1 \end{cases}$$

(i) Find $f(1)$

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(ii) Find x if $f(x) = 0$

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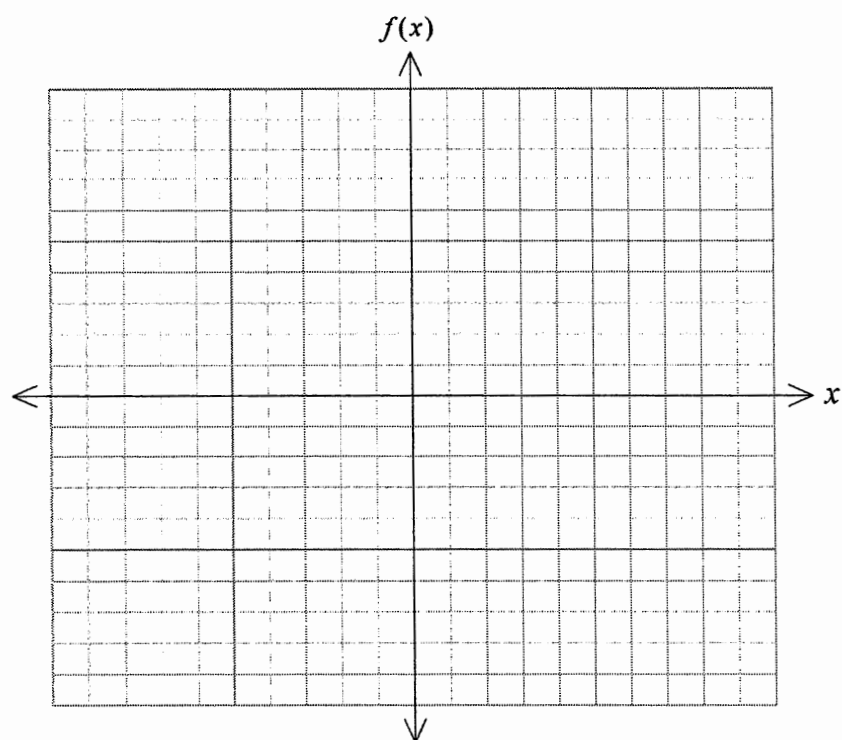
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(iii) Sketch the function showing all intercepts.

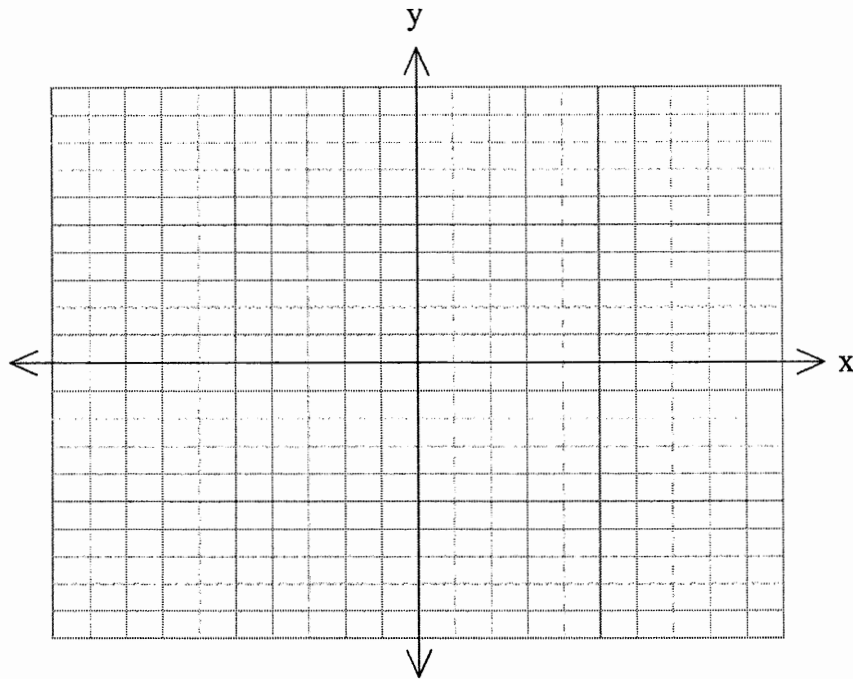
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End of Question 12

Question 13 (18 Marks)

- a) (i) Sketch the graphs of $f(x) = 2x - 2x^2$ and $g(x) = x - 1$ on the same number plane. 2



- (ii) Using your graphs from part (i), or otherwise solve the inequality 2

$$x - 1 < 2x - 2x^2.$$

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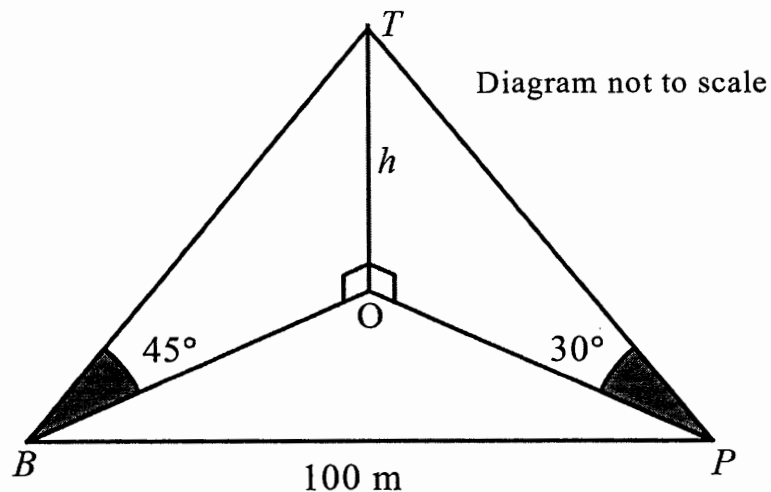
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Question 13 continued on the next page

- b) A surveyor stands at a point P , which is due east of the tower OT , of height h metres. The angle of elevation of the top of the tower T from P is 30° . The surveyor then walks 100 metres to point B , which is on a bearing of 150° from the foot of tower O . The angle of elevation of the top of the tower from B is now 45° .



- (i) Express the length of OP in terms of h .

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Question 13 continued on the next page

(ii) Show that $(100)^2 = h^2 + \frac{h^2}{\tan^2 30^\circ} - \frac{h^2}{\tan 30^\circ}$.

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(iii) Hence find the height of the tower. Answer correct to 1 decimal place. 1

[illegible]

Question 13 continued on the next page

- c) The following information shows a group of people's waist measurements and weights.

Waist (cm) x	72	67	85	96	80	90	98	105
Weight (kg) y	58	50	72	85	70	79	82	84

- (i) Calculate the correlation coefficient, r , for their waist and weight measurements 2 correct to 3 decimal places and hence describe the strength of the relationship.

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- (ii) Find the equation of the Least -Squares Regression Line. 1

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Question 13 continued on the next page

d) Given the function $f(x) = \ln(x^2 + 1)$.

(i) Find the domain of $f(x)$.

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(ii) Find any stationary point(s) and determine their nature.

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Question 13 continued on the next page

(iii) Find any point(s) of inflection.

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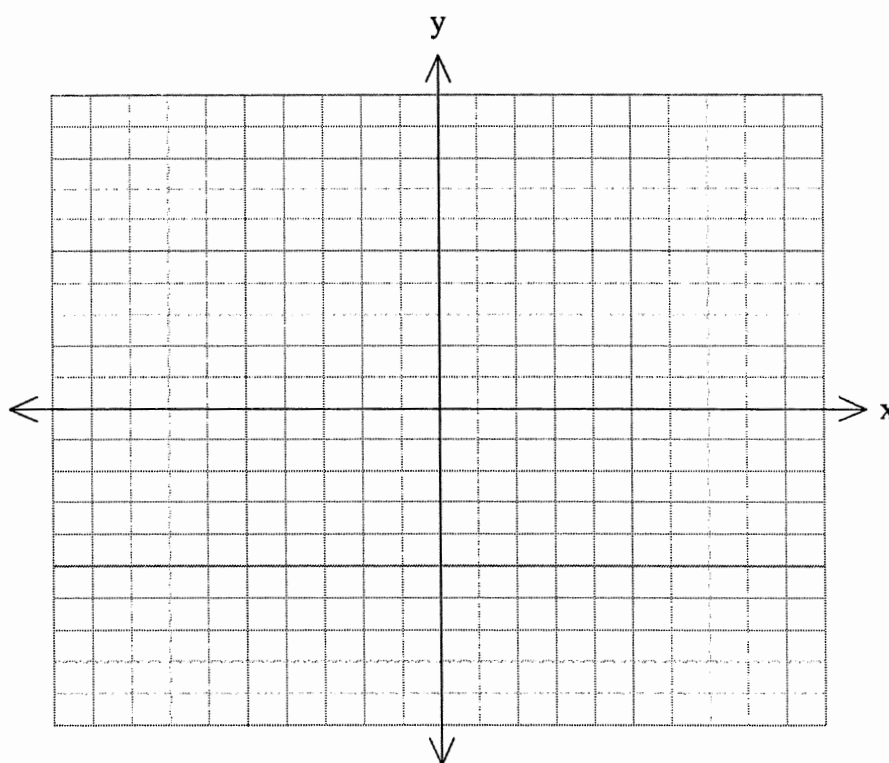
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(iv) Sketch the graph of $f(x) = \ln(x^2 + 1)$ showing all features from part (ii) and (iii).

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End of Question 13

Question 14 (14 marks)

- a) (i) Prove the following identity

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$$(1 + \tan x)^2 = 2 \tan x + \sec^2 x$$

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- (ii) Hence find the area bounded by $y = (1 + \tan x)^2$ and the x -axis between **3**

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}.$$

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Question 14 continued on the next page

b) Given $y = 2\sin\left(2x - \frac{\pi}{3}\right)$

(i) State the amplitude and period.

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(ii) Find the exact values of all intercepts of

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$y = 2\sin\left(2x - \frac{\pi}{3}\right)$ with the axes for $0 \leq x \leq \pi$.

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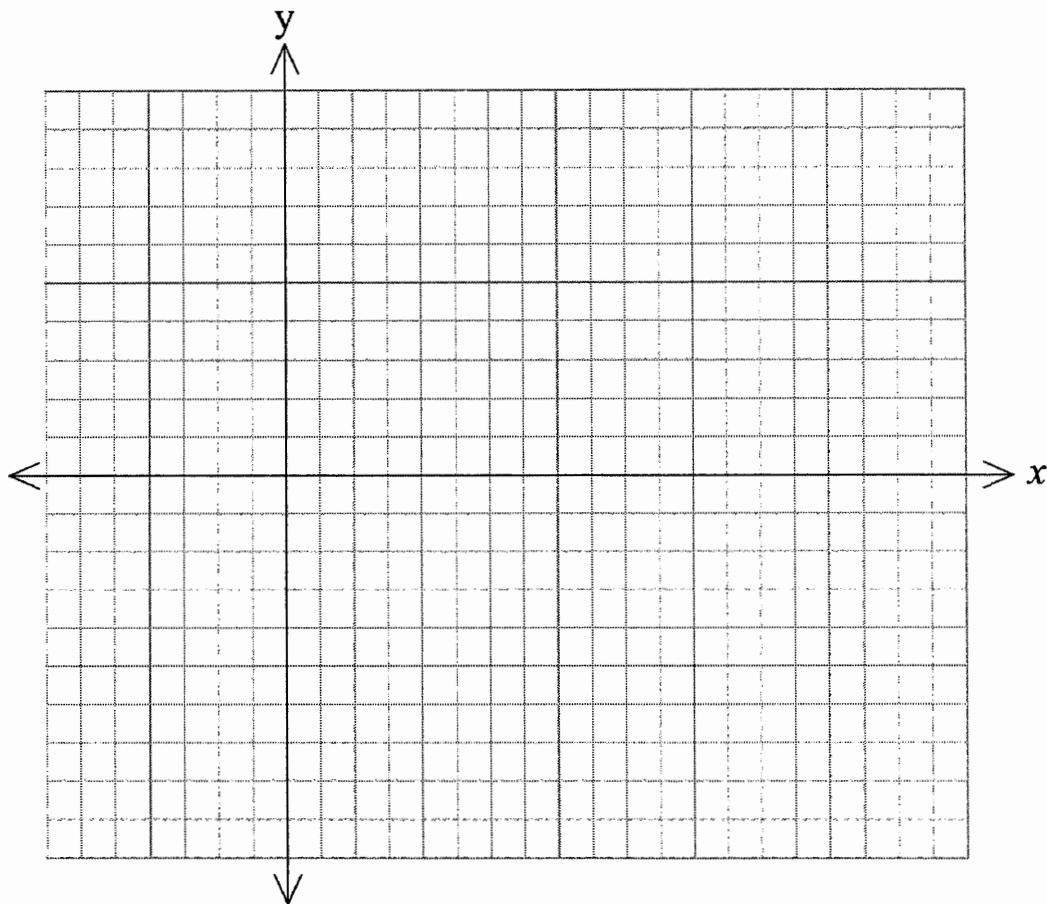
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Question 14 continued on the next page

- (iii) Hence sketch the graph of $y = 2\sin\left(2x - \frac{\pi}{3}\right)$ for $0 \leq x \leq \pi$, 2
showing all features from part (i) and (ii) and the global maximum and minimum.



Question 14 continued on the next page

c) A bag contains three red balls and four black balls. Two balls are selected at random without replacement from the bag.

Let X be the number of black balls drawn.

- (i) Fill in the following table and hence find exact value of $E(X)$. 2

x	0	1	2
$P(X = x)$			

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- (ii) Find $E(X^2)$ and hence find $\text{Var}(X)$ and standard deviation σ . 2

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End of question 14

Question 15 (16 marks)

a) The velocity v of a particle in metres per second is given by the formula

$$v = 5(1 + e^{-t}), \text{ where } t \text{ is the time in seconds.}$$

- (i) Find the initial velocity of the particle. 1

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- (ii) Is the particle ever stationary? Justify your answer. 1

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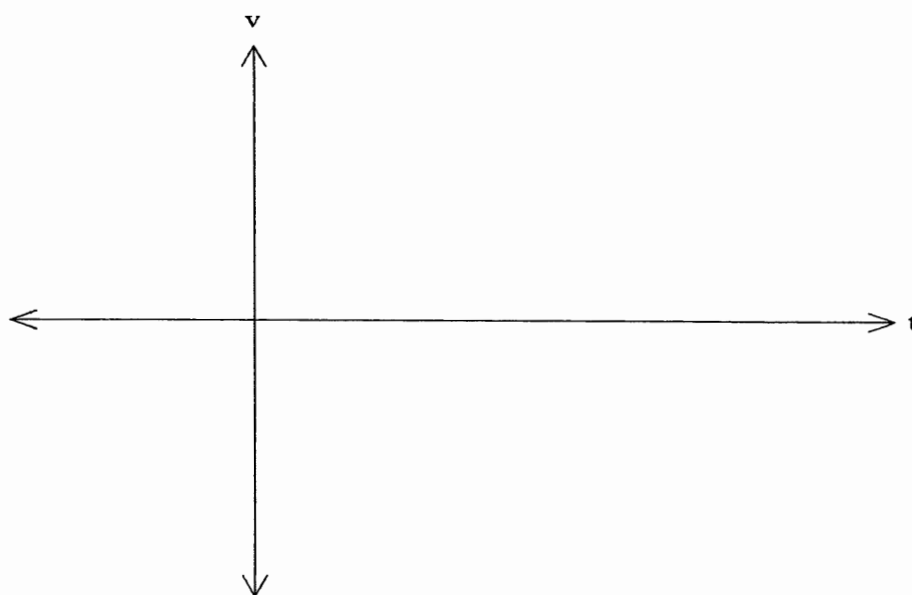
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- (iii) Sketch the graph of the velocity. 2



Question 15 continued on the next page

(iv) Find the total distance travelled by the particle in the first 5 seconds. 2

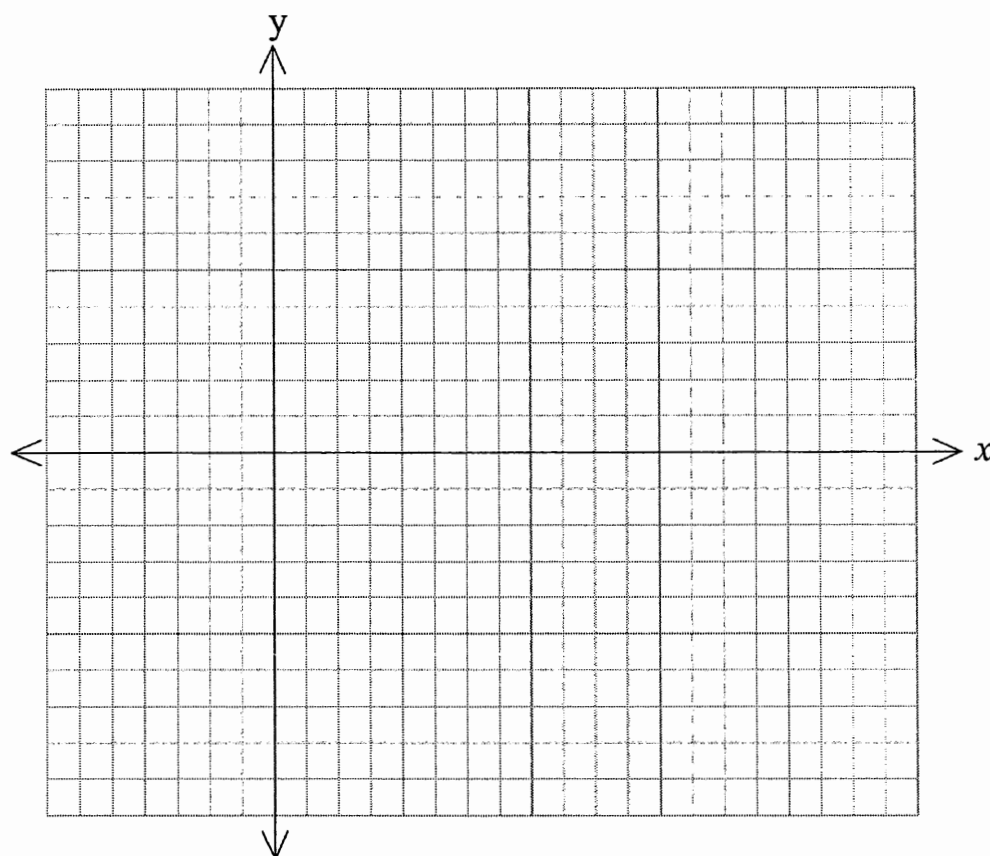
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Question 15 continued on the next page

b) The line $y = mx$ is a tangent to the curve $y = \ln(2x - 1)$ at a point P .

- (i) Sketch the line and the curve on the same diagram, clearly indicating the point P .

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Question 15 continued on the next page

- (ii) Show that the coordinates of P are $\left(\frac{2+m}{2m}, \frac{2+m}{2}\right)$. 2

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- (iii) Hence show that $2+m = \ln\left(\frac{4}{m^2}\right)$. 2

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Question 15 continued on the next page

c) Given the probability density function

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the cumulative distribution function $F(x)$. 2

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(ii) Hence find the median. 2

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End of question 15

Question 16 (14 marks)

- a) Michelle borrows \$450 000 to be repaid by regular monthly repayments of \$ M over a period of 25 years at 6% per annum reducible monthly. Interest is calculated and charged just before each repayment.

Let A_n be the amount owing after n –repayments.

- (i) Show that the expression for the amount owing after two repayments is 1

$$A_2 = 450\,000(1.005)^2 - M(1.005) - M.$$

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- (ii) Show that the amount owing after n –repayments is 2

$$A_n = 450\,000(1.005)^n - M \frac{(1.005)^n - 1}{0.005}.$$

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Question 16 continued on the next page

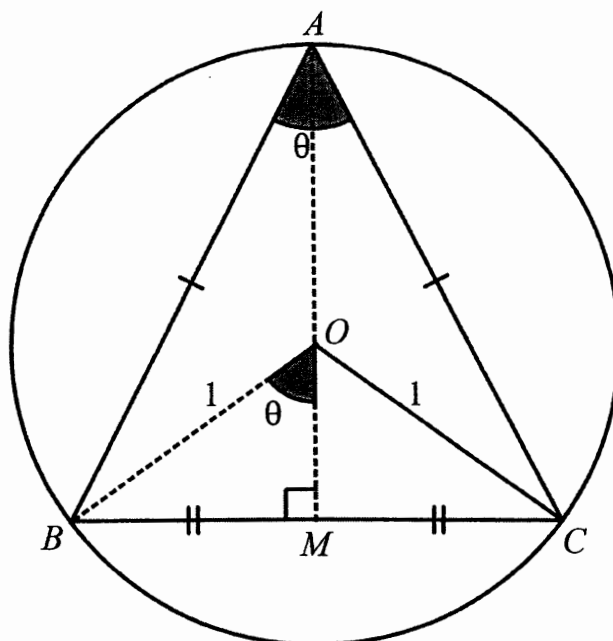
(iii) Calculate the amount of each regular monthly repayment.

2

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Question 16 continued on the next page

- b) An isosceles triangle $\triangle ABC$ is inscribed within a unit circle centred at O , as shown in the diagram below. Let M be the midpoint of BC , $\angle BAC = \theta$ and $\angle BOM = \theta$.



- (i) Show that the area of $\triangle ABC$ is $A = \sin\theta(1 + \cos\theta)$.

2

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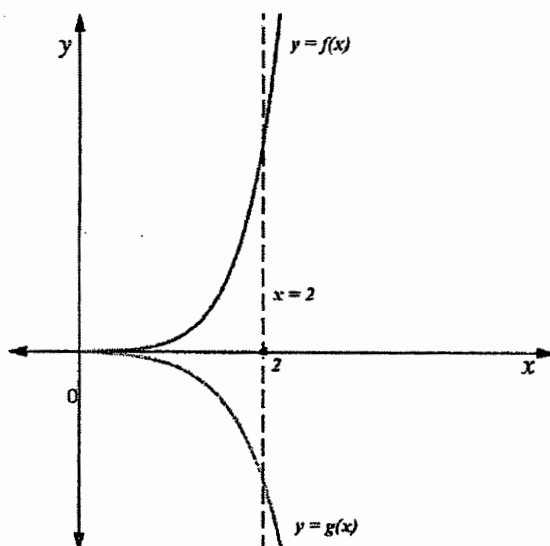
Question 16 continued on the next page

- (ii) Hence prove that the area of the isosceles triangle $\triangle ABC$ is maximum when it is equilateral.

[illegible]

Question 16 continued on the next page

- c) The graph of $f(x) = x^2 e^{kx}$ and $g(x) = -\frac{2xe^{kx}}{k}$ and the line $x = 2$ is drawn below, where k is a positive constant. $f(x) = g(x)$ at only one point, that is at $(0, 0)$.



Let A be the area of the region bounded by the curve $y = f(x)$, $y = g(x)$ and the line $x = 2$.

- (i) Write down a definite integral that gives the value of A .

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- (ii) The function $f(x)$ from part (i) is given by $f(x) = x^2 e^{kx}$, where k is a positive constant. Show that $f'(x) = xe^{kx}(kx + 2)$.

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Question 16 continued on the next page

(iii) Using the results of part (i) and (ii), or otherwise, find the value of k such that 2

$$A = \frac{16}{k}.$$

[illegible]

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End of Exam

Answers - Multiple Choice (2020)
Yr. 12 TRIAL - Maths Advanced

(1) $\int \tan^2 x \, dx = \int \sec^2 x - 1 \, dx$ (B)
 $= \tan x - x + C$

(2) $\frac{d}{dx} \log_e \frac{4x^2-9}{2x-3} = \frac{d}{dx} \log_e \frac{(2x-3)(2x+3)}{(2x-3)}$ (B)
 $= \frac{d}{dx} \log_e (2x+3) = \frac{2}{2x+3}$

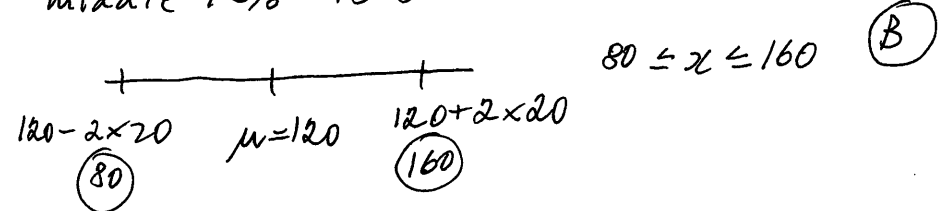
(3) $f'(x) = \frac{x}{e^{x^2-8}} \therefore f(x) = \int \frac{x}{e^{x^2-8}} \, dx$
 $f(x) = \int x \cdot (e^{x^2-8})^{-1} \, dx = \int x e^{8-x^2} \, dx$
 $= -\frac{1}{2} \int \underbrace{2x \cdot e^{8-x^2}}_{g'(x)} \underbrace{1}_{g(x)} \, dx = -\frac{1}{2} e^{8-x^2} + C$ (D)
but C can be any constant $\therefore$

(4) curve is decreasing $\therefore \frac{dy}{dx} < 0$
curve is concave up $\therefore \frac{d^2y}{dx^2} > 0$ } (D)

(5) $\mu = \bar{x} = 55$ $(+3\sigma)$ $z=3$ $\therefore \text{score} = 55 + 3 \times 6 = 73$ (B)

(6) Area = 34 = $\left| \int_1^2 k(x-2)^3 \, dx \right| + \int_2^4 k(x-2)^3 \, dx$
 $34 = k \left| \int_1^2 (x-2)^3 \, dx \right| + k \int_2^4 (x-2)^3 \, dx$
 $34 = k \left| \left[\frac{(x-2)^4}{4} \right]_1^2 \right| + k \left[\frac{(x-2)^4}{4} \right]_2^4$
 $34 = k \left| 0 - \frac{1}{4} \right| + k \left[\frac{16}{4} - 0 \right]$
 $34 = k \left(\frac{1}{4} + 4 \right)$ (B)
 $\therefore k = 8$

(7) $\mu = 120$ $\sigma = 20$ ($\text{Var}(T) = \sigma^2$)
middle 95% is 2σ around μ .



(8) $I \doteq \frac{h}{2} (\text{first value} + \text{last value})$
 $= \frac{1}{2} \left(\frac{\ln 1}{2} + \frac{\ln 2}{2} \right) = \frac{1}{4} \ln 2 = 0.17328\dots$ (smaller)
Exact $I = \frac{1}{2} (\ln 2)^2 = 0.2402265$
 $\therefore \% = \frac{\text{approx. } I - \text{Exact } I}{\text{Exact } I} = \frac{\frac{1}{4} \ln 2 - \frac{1}{2} (\ln 2)^2}{\frac{1}{2} (\ln 2)^2} \times 100$
 $\% = 27.865\% \doteq 28\%$ smaller (A)

(9) $f(-x) = \frac{-x}{(-x)^2-5} = \frac{-x}{x^2-5} = -f(x) \therefore \text{odd}$
by horiz. line test $\therefore$ many to one } (D)

(10) $M = 9t^2 - t^3$
 $\frac{dM}{dt} = 18t - 3t^2 = 3t(6-t)$
 $t=3 \dots$ (D)

Section II- Extended Response

Attempt Questions 11-16.

Allow about 75 minutes for this section.

Question 11(14 Marks)

a) Differentiate the following

(i) $y = (4x - 5)(4x + 5) = 16x^2 - 25$

$$\frac{dy}{dx} = 32x$$

Marks 1

1 - correct soln

OR $\frac{dy}{dx} = 4(4x+5) + (4x-5) \times 4$
 $= 16x + 20 + 16x - 20 = 32x$

(ii) $y = \sin^2 x$

2

$$\frac{dy}{dx} = 2 \sin x \cdot \cos x$$

2 - correct soln.

1 - correctly diff.

$\sin x$

1 - $\frac{dy}{dx} = 2 \sin x \cos x$

b) In AP, $T_3 = 5$ and $T_{10} = 26$.
Find the sum of S_{14} .

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AP: $T_3 = 5$ $T_{10} = 26$

$$T_3 = 5 = a + 2d$$

2 - correct soln.

$$T_{10} = 26 = a + 9d$$

1 - correctly finds

$$21 = 7d \therefore d = 3, a = -1$$

a or d

$$\therefore S_{14} = \frac{14}{2} (2a + (14-1)d) = 259$$

1 - applies S_n for A.P correctly

Question 11 continued on the next page

c) Evaluate

$$\int_1^4 5(9x - 4)^4 dx$$

$$= \left[5 \times \frac{(9x-4)^5}{5 \times 9} \right]_1^4$$

2 - correct soln.

1 - correct integral

$$= \frac{1}{9} \left[(9x-4)^5 \right]_1^4$$

1 - correct answer from incorrect integral

$$= \frac{1}{9} \left[(9(4)-4)^5 - (9-4)^5 \right]$$

$$= \frac{1}{9} \left[32^5 - 5^5 \right] = 3727923$$

d) $e^{2x} + 3e^x - 10 = 0$

2 - correct soln.

let $m = e^x$

1 - correctly reduces to quadratic eqn & solves it correctly

$$m^2 + 3m - 10 = 0$$

$$(m+5)(m-2) = 0$$

$$m = -5$$

$$m = +2$$

$$e^x = -5$$

$$e^x = 2$$

no solns.

$$x = \ln 2$$

$\therefore$ solution $x = \ln 2$ (only)

Question 11 continued on the next page

e)

(i) Show that $\frac{d}{dx}(\sec^2 x) = 2 \tan x \sec^2 x$

2

$$\begin{aligned}
 \text{LHS} &= \frac{d}{dx}(\sec^2 x) = \frac{d}{dx}\left(\frac{1}{\cos^2 x}\right) \quad \begin{array}{l} 2 - \text{correct soln.} \\ 1 - \text{differentiates} \\ \text{correctly } \sec^2 x \end{array} \\
 &= \frac{d}{dx}(\cos^{-2} x) = -2 \cos^{-3} x \cdot (-\sin x) \\
 &= \frac{2 \sin x}{\cos^3 x} = \frac{2 \sin x}{\cos x} \times \frac{1}{\cos^2 x} \quad \begin{array}{l} 1 - \text{applies trig.} \\ \text{identities correctly} \end{array} \\
 &= 2 \tan x \cdot \sec^2 x \\
 &= \text{RHS} \therefore \text{shown}
 \end{aligned}$$

(ii) Hence find $\int \tan x \sec^2 x dx$

1

$$\begin{aligned}
 \text{from (i)} \quad \frac{d}{dx}(\sec^2 x) &= 2 \tan x \sec^2 x \quad 1 - \text{correct soln.} \\
 \therefore \sec^2 x &= \int 2 \tan x \sec^2 x dx \quad - \text{ignore } +C \\
 \therefore \frac{1}{2} \sec^2 x &= \int \tan x \sec^2 x dx \\
 \therefore \int \tan x \sec^2 x dx &= \frac{1}{2} \sec^2 x + C \\
 \text{OR } &= \frac{1}{2} (1 + \tan^2 x) + C
 \end{aligned}$$

Question 11 continued on the next page

f) Given a function $f(x) = \begin{cases} 6x - 6x^2 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$ (i) Show that $f(x)$ represents probability density function.

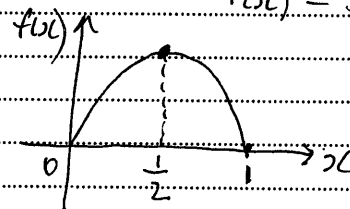
2

$$\begin{aligned}
 f(x) \text{ represents PDF if} & \quad 2 - \text{correct soln.} \\
 \bullet \int_0^1 f(x) dx &= 1 \text{ and } f(x) \geq 0 \quad \begin{array}{l} 1 - \text{finds } \int f(x) dx \\ \text{on domain} \end{array} \\
 \therefore \int_0^1 (6x - 6x^2) dx & \quad \text{true} \quad \text{correctly} \\
 &= \left[3x^2 - 2x^3 \right]_0^1 \\
 &= [(3-2) - 0] = 1 \therefore \text{Yes it's PDF}
 \end{aligned}$$

(ii) Find the mode of the probability density function $f(x)$

1

$$\begin{aligned}
 \text{By sketching } f(x) &= 6x - 6x^2 \\
 f(x) &= 6x(1-x)
 \end{aligned}$$



Mode is $x = \frac{1}{2}$ with the highest value of $f(x)$

1 - correct soln.

End of Question 11

Question 12 (3 Marks)

- a) Find the value(s) of m such that $y = 2x + m$ is a tangent to the parabola

$$y = 2x^2 + 6x - 5.$$

(2)

2 + correct soln.

$y = 2x + m$
 $2 \Rightarrow$ gradient of tangent

1 - finds point of contact by using calculus

$\therefore y' = 4x + 6$ where $m = 2$
 $\therefore 2 = 4x + 6$
 $-1 = x \quad \therefore y = 2(-1)^2 + 6(-1) - 5$
 $y = -9$
 $\therefore$ pt. of contact $(-1, -9)$

1 - finds Δ correctly by using simult. eqns.

Sub. $(-1, -9)$ into $y = 2x + m$
 $-9 = 2(-1) + m$
 $\therefore m = -7$

1 - finds gradient function correctly & x-coord. of pt. of contact

OR by simultaneous eqn.

$2x + m = 2x^2 + 6x - 5$

1 - creates quadr. eqn. correctly by solving simult. eqns, & attempts to solve $\Delta = 0$

$2x^2 + 4x - 5 - m = 0$
 $\Delta = 0$ (since tangent, only one solution)
 $0 = b^2 - 4ac$
 $0 = 4^2 - 4(2)(-5 - m)$
 $0 = 16 + 40 + 8m$
 $m = -7$

Question 12 continued on the next page

- b) Angela guesses three questions in her multiple choice test, which has four options per question. Find the probability that Angela gets

1/4 C 1/4 C 1/4 C
 1/4 Correct (C) 3/4 IC 3/4 IC 3/4 IC
 3/4 Incorrect (IC) 1/4 C 1/4 C 1/4 C
 3/4 IC 3/4 IC 3/4 IC

- i) Only one correct

1 - correct soln.

$$P = P(Ci\bar{i}) + P(\bar{i}Ci\bar{i}) + P(\bar{i}\bar{i}Ci)$$

$$= \left(\frac{1}{4}\right)^2 \times \left(\frac{3}{4}\right) + \left(\frac{1}{4}\right) \times \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^2 \times \left(\frac{1}{4}\right)$$

$$= 3 \times \left(\frac{1}{4}\right) \times \left(\frac{3}{4}\right)^2 = \frac{27}{64}$$

- (ii) At least one correct

1 - correct soln.

$$P(\text{at least one correct})$$

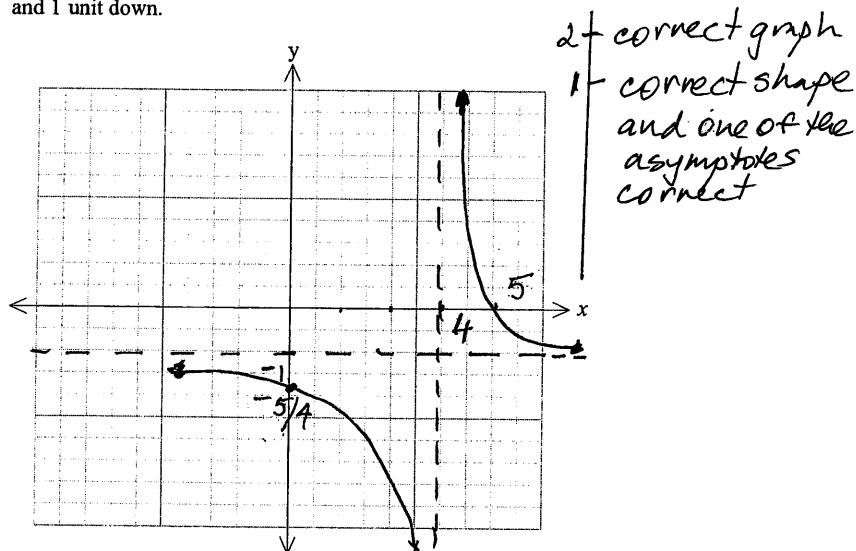
$$= 1 - P(\text{no correct})$$

$$= 1 - \left(\frac{3}{4}\right)^3 = \frac{37}{64}$$

Question 12 continued on the next page

c)

- (i) Sketch the hyperbola by shifting $y = \frac{1}{x-1}$ horizontally 3 units to the right and 1 unit down. 2



- (ii) State the equation of the shifted hyperbola, then find all the intercepts of the shifted hyperbola with the axes and mark them on your graph in part (i). 2

$$y = \frac{1}{x-4} - 1$$

2- correct soln.
1- correct eqn.

$$x=0 \therefore y = \frac{1}{0-4} - 1 = -\frac{1}{4} \text{ or } -\frac{5}{4}$$

1- one of the intercepts correct and labeled on the graph

$$\therefore y\text{-int } (0, -\frac{5}{4})$$

$$y=0 \therefore 0 = \frac{1}{x-4} - 1$$

$$1 = \frac{1}{x-4} \therefore x\text{-int } (5, 0)$$

$$x-4=1 \therefore x=5$$

Question 12 continued on the next page

- (d) Consider the piece-wise defined function.

$$f(x) = \begin{cases} x^2 - 1 & x \leq 1 \\ 4 - x^2 & x > 1 \end{cases}$$

- (i) Find $f(1)$

$$f(1) = 1^2 - 1 = 0$$

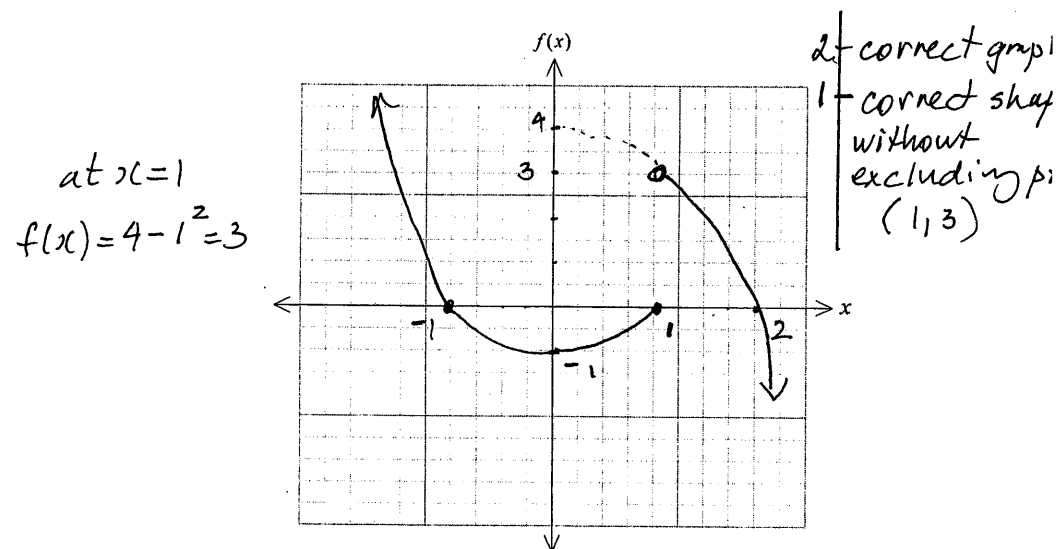
1- correct soln.

- (ii) Find x if $f(x) = 0$

$$f(x) = 0 \begin{cases} x^2 - 1 = 0 \therefore x = \pm 1 \\ 4 - x^2 = 0 \therefore x = \pm 2 \end{cases}$$

2- correct answers
1- all 4 solns.
correct with one excluding $x = -1$
1- $x = \pm 1$

- (iii) Sketch the function showing all intercepts. 2

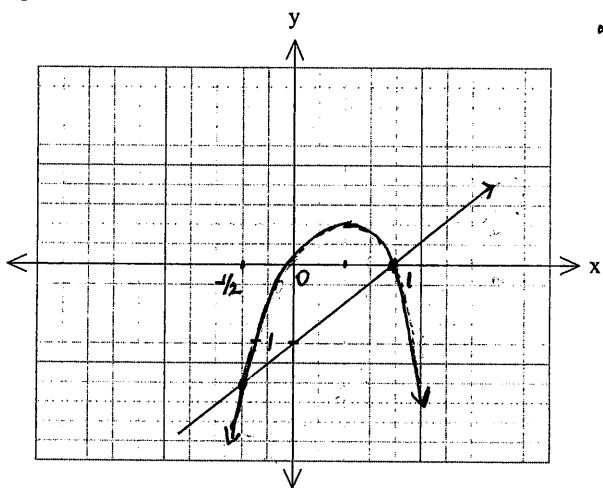


End of Question 12

Question 13 (18 Marks)

- a) (i) Sketch the graphs of $f(x) = 2x - 2x^2$ and $g(x) = x - 1$ on the same number plane.

$$f(x) = 2x(1-x)$$



2 - both graphs correct
1 - correct graph of $g(x)$ with correct intercept
1 - $f(x)$ correct graph with correct x-intercepts

- (ii) Using your graphs from part (i), or otherwise solve the inequality

$$x - 1 < 2x - 2x^2$$

$f(x) \cap g(x) = \text{pts. of intersection}$

$$x - 1 = 2x - 2x^2$$

$$2x^2 - x + 1 = 0$$

$$(2x+1)(x-1) = 0$$

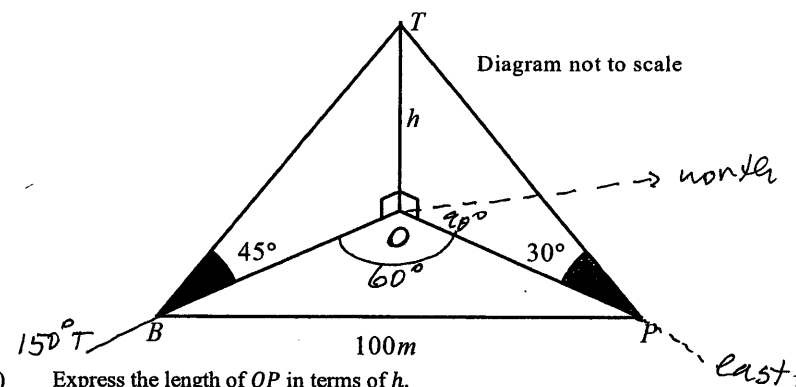
$$x = -\frac{1}{2} \quad x = 1$$

$\therefore$ answer graphically (line below parabola)
 $-\frac{1}{2} < x < 1$

2 - correct solus.
1 - finds pts. of intersection
 $f(x) \cap g(x)$ correctly

Question 13 continued on the next page

- b) A surveyor stands at a point P , which is due east of the tower OT , of height h metres. The angle of elevation of the top of the tower T from P is 30° . The surveyor then walks 100 metres to point B , which is on a bearing of 150° from the foot of tower O . The angle of elevation of the top of the tower from B is now 45° .



- (i) Express the length of OP in terms of h .

from $\triangle OPT$

$$\tan 30^\circ = \frac{h}{OP}$$

$$OP = \frac{h}{\tan 30^\circ}$$

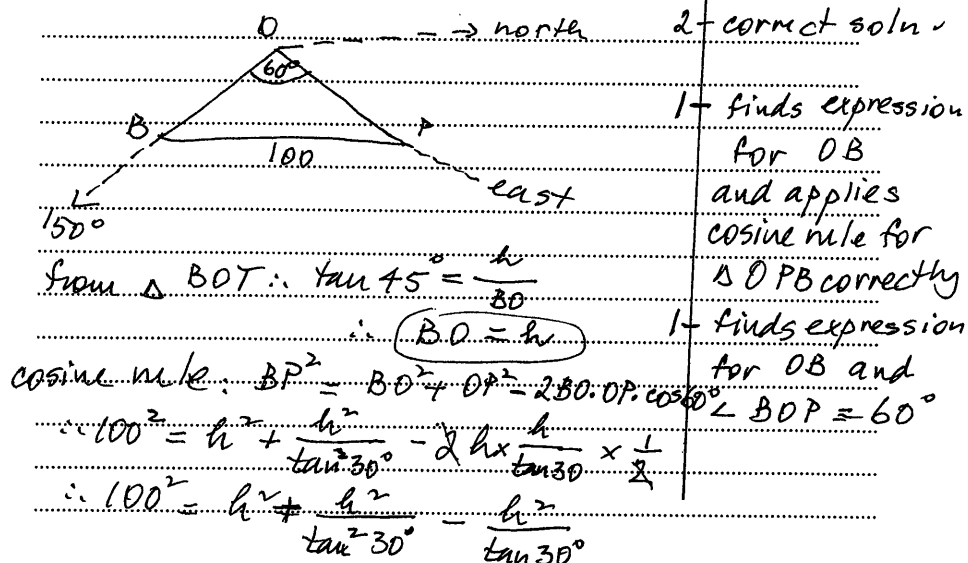
$$\text{OR } OP = \sqrt{3}h$$

1 - correct expression with $\tan 30^\circ$
1 - correct answer $OP = \sqrt{3}h$

Question 13 continued on the next page

(ii) Show that $(100)^2 = h^2 + \frac{h^2}{\tan^2 30^\circ} - \frac{h^2}{\tan 30^\circ}$

2



(iii) Hence find the height of the tower. Answer correct to 1 decimal place.

1

$$100^2 = h^2 \left(1 + \frac{1}{\tan^2 30^\circ} - \frac{1}{\tan 30^\circ} \right)$$

$$h^2 = \frac{100^2}{\left(1 + \frac{1}{\tan^2 30^\circ} - \frac{1}{\tan 30^\circ} \right)}$$

$$h^2 = 4409.269852$$

$$\therefore h = 66.4 \text{ m (1dp)}$$

1 - correct answer

(ignore rounding)

c) The following information shows a group of people's waist measurements and weights.

Waist (cm) x	72	67	85	96	80	90	98	105
Weight (kg) y	58	50	72	85	70	79	82	84

(i) Calculate the correlation coefficient, r, for their waist and weight measurements and hence describe the strength of the relationship.

2

$$r = 0.9592$$

(calculator)

$\therefore$ strong correlation positive

2 - correct soln.

1 - correct r

1 - from their "r" correct conclusion for strength of the relationship

(ii) Find the equation of the Least-Squares Regression Line.

1

from calculator

$$A = -8.2368$$

$$B = 0.93203$$

1 - correct solns.

$$y = A + Bx$$

$$y = -8.2368 + 0.93203x$$

Question 13 continued on the next page

d) Given the function $f(x) = \ln(x^2 + 1)$.

(i) Find the domain of $f(x)$.

1

$$x^2 + 1 > 0$$

which is always

$\therefore$ Domain = all real x

1- correct solns.

(ii) Find any stationary point(s) and determine their nature.

2

$$f'(x) = \frac{2x}{x^2 + 1}$$

2- correct solns.

$$0 = \frac{2x}{x^2 + 1}$$

1- finds stationary point correctly

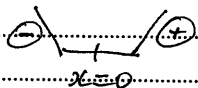
$$2x = 0, \quad y = \ln(0+1) = 0$$

1- determines the nature of st. point correctly

$\therefore (0, 0)$ is stationary point

Nature by f'' or table

x	-1	0	1
f''	-1	0	1



$\therefore (0, 0)$ is minimum turning pt.

(or) $f''(x) = \frac{2 - 2x^2}{(x^2 + 1)^2}$

$$f''(0) = 2 > 0 \quad \therefore (0, 0) \text{ is min. t.p.}$$

Question 13 continued on the next page

(iii) Find any point(s) of inflection.

2

$$f''(x) = \frac{2(x^2 + 1) - 2x(2x)}{(x^2 + 1)^2}$$

2- correct solns.

$$f''(x) = \frac{2 - 2x^2}{(x^2 + 1)^2} = 0$$

1- solves for $f'' = 0$ correctly

$$2 - 2x^2 = 0 \quad \therefore x = \pm 1$$

1- justifies pt. of inflexion correctly

$\therefore (1, \ln 2), (-1, \ln 2)$ possible pts. of inflexion

x	-2	-1	0	1	2
f''	$-\frac{6}{25}$	0	2	0	$-\frac{6}{25}$

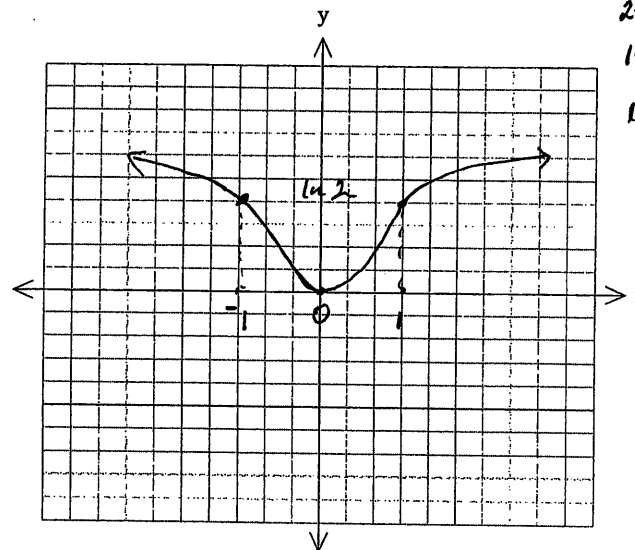
concavity changes at $x = \pm 1$

$\therefore (1, \ln 2), (-1, \ln 2)$ are points of inflexion

(iv) Sketch the graph of $f(x) = \ln(x^2 + 1)$ showing all features from

2

part (ii) and (iii).



2- correct graph showing all features
1- correct shape
1- shows correctly features from part (ii) & (iii)

End of Question 13

Question 14 (14 marks)

- a) (i) Prove the following identity

$$(1 + \tan x)^2 = 2 \tan x + \sec^2 x$$

$$\begin{aligned} \text{LHS} &= (1 + \tan x)^2 \\ &= 1 + 2 \tan x + \tan^2 x \\ &= \underbrace{1 + \tan^2 x} + 2 \tan x \\ &= \sec^2 x + 2 \tan x \\ &= \text{RHS} \end{aligned}$$

1 - correct solns

- (ii) Hence find the area bounded by $y = (1 + \tan x)^2$ and the x-axis between

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

since $(1 + \tan x)^2 \geq 0$ for $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

$$\text{Area} = \int_{-\pi/4}^{\pi/4} (1 + \tan x)^2 dx$$

$$= \int_{-\pi/4}^{\pi/4} 2 \tan x + \sec^2 x dx$$

$$= \left[-2 \ln |\cos x| + \tan x \right]_{-\pi/4}^{\pi/4}$$

$$= \left[(-2 \ln |\cos \frac{\pi}{4}| + 1) - (-2 \ln |\cos \frac{-\pi}{4}| + (-1)) \right]$$

$$= 2$$

2 - correct solns.
1 - correctly integrates $\tan x$
1 - correctly uses part (i)
1 - correctly evaluates the definite integral

Question 14 continued on the next page

b) given $y = 2 \sin \left(2x - \frac{\pi}{3} \right)$

- (i) State the amplitude and period.

amplitude = 2

period $T = \frac{2\pi}{2} = \pi$

2 - correct solns.

1 - correct amplitude

1 - correct period

- (ii) Find all intercepts of $y = 2 \sin \left(2x - \frac{\pi}{3} \right)$ with the axes for $0 \leq x \leq \pi$

y-int: $x = 0$

$$y = 2 \sin \left(2(0) - \frac{\pi}{3} \right)$$

$$y = -2 \frac{\sqrt{3}}{2} = -\sqrt{3}$$

$\therefore (0, -\sqrt{3})$ y-int.

x-int: $y = 0$

$$0 = 2 \sin \left(2x - \frac{\pi}{3} \right)$$

$$2x - \frac{\pi}{3} = 0, \pi, 2\pi$$

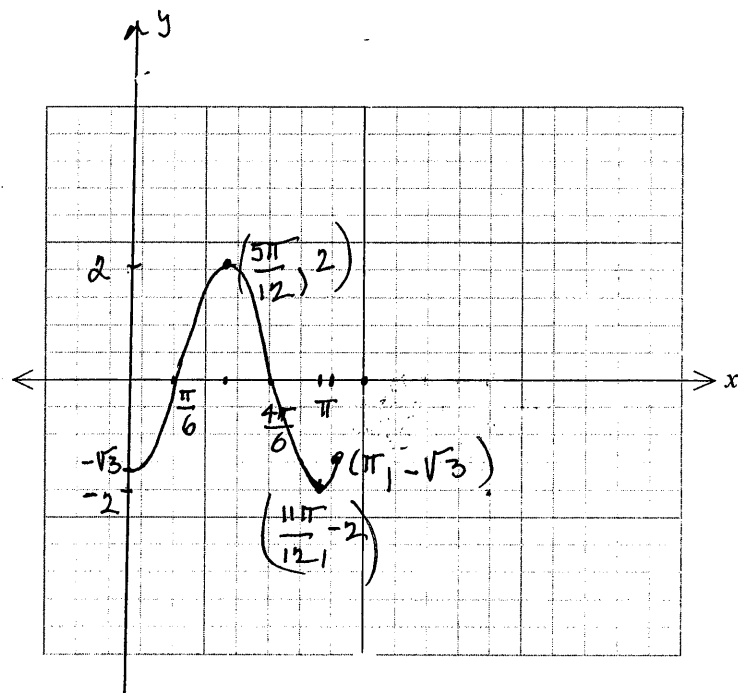
$$\therefore x = \frac{\pi}{6}, \frac{4\pi}{6}, \frac{7\pi}{6}$$

out of domain

$\therefore \left(\frac{\pi}{6}, 0 \right), \left(\frac{2\pi}{3}, 0 \right)$ are x-int.

Question 14 continued on the next page

- (iii) Hence sketch the graph of $y = 2\sin\left(2x - \frac{\pi}{3}\right)$ for $0 \leq x \leq \pi$ showing all features from part (i) and (ii) and global maximum and minimum. 2



- 2 - correct graph
1 - correct shape and x, y-intercepts
1 - showing correct Max/Min

- c) A bag contains three red balls and four black balls. Two balls are selected at random without replacement from the bag.

Let X be the number of black balls drawn. exact value of

- (i) Fill in the following table and hence find $E(X)$. 2

x	0	1	2
$P(X = x)$	RR $1/7$	RB or BR $4/7$	BB $2/7$

Handwritten calculations for probabilities and expected value:

$$P(0) = \frac{3}{7} \times \frac{2}{6} = \frac{1}{7}$$

$$P(1) = 2 \times \frac{3}{7} \times \frac{4}{6} = \frac{4}{7}$$

$$P(2) = \frac{4}{7} \times \frac{3}{6} = \frac{2}{7}$$

2 - correct soln.
1 - Finds correct $E(X) = \mu$
from incorrect table of values

$$E(X) = \sum x p(x) = 0 \times \frac{1}{7} + 1 \times \frac{4}{7} + 2 \times \frac{2}{7} = \frac{8}{7}$$

- (ii) Find $E(X^2)$ and hence find $\text{Var}(X)$ and σ 2

$$E(X^2) = \sum x^2 p(x) = 0 \times \frac{1}{7} + 1^2 \times \frac{4}{7} + 2^2 \times \frac{2}{7}$$

$$\therefore E(X^2) = \frac{12}{7}$$

$$\text{Var}(X) = E(X^2) - \mu^2$$

$$= \frac{12}{7} - \left(\frac{8}{7}\right)^2 = \frac{20}{49}$$

$$\text{Var}(X) = \frac{20}{49} \therefore \sigma = 0.638976\dots$$

$$\sigma = \sqrt{\frac{20}{49}} = \frac{\sqrt{20}}{7} = \frac{2\sqrt{5}}{7} \therefore \sigma = 0.64396$$

End of question 14

Question 14 continued on the next page

Question 15 (16 marks)

a) The velocity v of a particle in metres/seconds is given by the formula

$$v = 5(1 + e^{-t}), \text{ where } t \text{ is time in seconds.}$$

(i) Find the initial velocity of the particle.

1

$$t = 0$$

$$v = 5(1 + e^{-0}) = 10 \text{ m/s}$$

1 - correct soln.

(ii) Is the particle ever stationary? Justify your answer.

1

$$v = 0$$

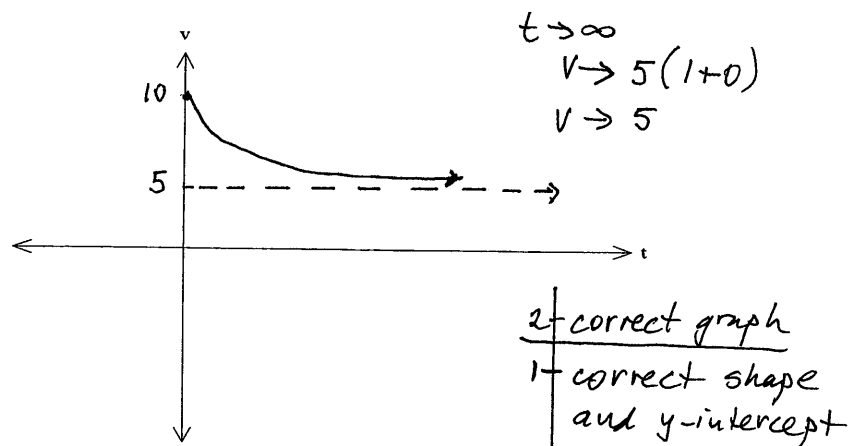
$$0 = 5(1 + e^{-t})$$

$$\therefore \text{no soln.} \therefore v \neq 0$$

1 - correct soln.

(iii) Sketch the graph of the velocity.

2



Question 15 continued on the next page

(iv) Find the total distance travelled by the particle in the first 5 seconds.

2

$$d = \int_0^5 5(1 + e^{-t}) dt$$

2 - correct solns.
1 - correct integration

$$= 5 \left[t - e^{-t} \right]_0^5$$

$$= 5 \left[5 - e^{-5} - (0 - e^0) \right]$$

$$= 5(5 - e^{-5} + 1)$$

$$= 30 - 5e^{-5} \text{ (metres) exact}$$

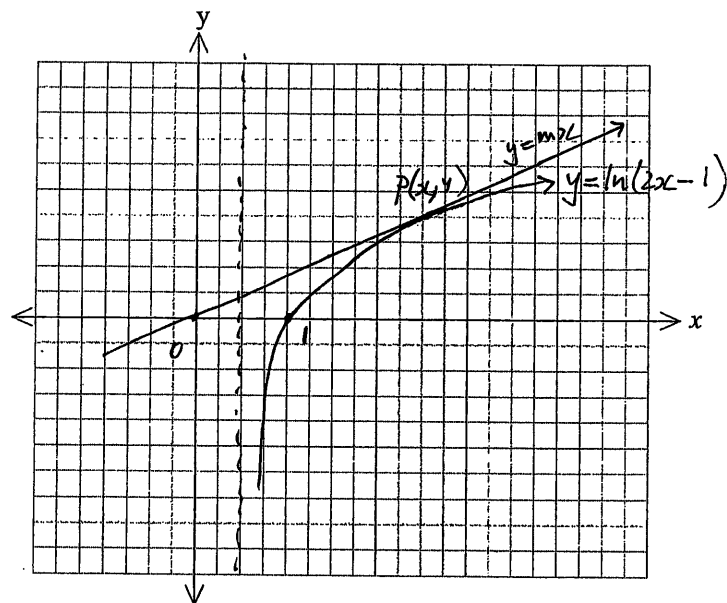
(OR) $d = 29.97 \text{ (2d.p.)}$

Question 15 continued on the next page

b) The line $y = mx$ is a tangent to the curve $y = \ln(2x - 1)$ at a point P .

(i) Sketch the line and the curve on the same diagram, clearly indicating the point P .

2



$$y = \ln(2x - 1)$$

$$2x - 1 > 0$$

$$x > \frac{1}{2}$$

$$2x - 1 = 1$$

$$x = 1$$

(x-int)

$$y = mx \rightarrow \text{passing through } (0, 0)$$

2- correct graphs

1- correct log. graph

1- correct graph

of the line passing through (0, 0) & clearly indicating point of contact $P(x, y)$.

Question 15 continued on the next page

(ii) Show that the coordinates of P are $\left(\frac{2+m}{2m}, \frac{2+m}{2}\right)$.

2

$$y = \ln(2x - 1)$$

$$\therefore y' = \frac{2}{2x - 1}$$

if $y = mx$ is a tangent to $y = \ln(2x - 1)$ at point $P(x, y)$

$$\therefore m = \frac{2}{2x - 1}$$

$$2x - 1 = \frac{2}{m}$$

$$2x = \frac{2}{m} + 1 \therefore x_p = \frac{1}{m} + \frac{1}{2} = \frac{2+m}{2m}$$

sub. x_{coord} into $y = mx$

$$\therefore y_p = mx \cdot \frac{2+m}{2m} = \frac{2+m}{2}$$

(iii) Hence show that $2 + m = \ln\left(\frac{4}{m^2}\right)$.

2

since $P\left(\frac{2+m}{2m}, \frac{2+m}{2}\right)$ (part ii)

and P lies on $y = \ln(2x - 1)$

$\therefore$ coord. of P satisfy equation

$\therefore$ sub. in $y = \ln(2x - 1)$

$$\frac{2+m}{2} = \ln\left(2\left(\frac{2+m}{2m}\right) - 1\right)$$

$$\frac{2+m}{2} = \ln\left(\frac{2+m}{m} - 1\right) = \ln\left(\frac{2}{m} + \frac{m}{m} - 1\right)$$

$$\frac{2+m}{2} = \ln\left(\frac{2}{m} + 1 - 1\right) = \ln\left(\frac{2}{m}\right)$$

$$\therefore 2+m = 2 \ln\left(\frac{2}{m}\right) = \ln\left(\frac{2}{m}\right)^2 = \ln\frac{4}{m^2}$$

Question 15 continued on the next page

$$\therefore 2+m = \ln\left(\frac{4}{m^2}\right) \therefore \text{shown}$$

c) Given the probability density function

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the cumulative distribution function $F(x)$.

2

$$F(x) = \int_0^x 2e^{-2x} dx$$

$$F(x) = 2x - \frac{1}{2} [e^{-2x}]_0^x$$

$$F(x) = -[e^{-2x} - e^0]$$

$$\therefore F(x) = -e^{-2x} + 1$$

2 - correct solns.
1 - correctly integrates $f(x)$

(ii) Hence find the median.

2

let m be median

$$\therefore F(m) = \frac{1}{2}$$

$$\frac{1}{2} = -e^{-2x} + 1$$

$$e^{-2x} = 1 - \frac{1}{2}$$

$$\ln(e^{-2x}) = \ln \frac{1}{2}$$

$$-2x = \ln \frac{1}{2}$$

$$x = -\frac{1}{2} \ln \frac{1}{2} \quad (\text{median})$$

$$\therefore \text{median} = -\frac{1}{2} \ln \frac{1}{2} \text{ or } \ln \sqrt{2} \text{ or } 0.347$$

$$\text{OR } -\frac{1}{2} (-\ln 2) = \frac{1}{2} \ln 2$$

2 - correct solns.
1 - equates $F(x) = \frac{1}{2}$
and shows significant progress to find the value of the median.

End of question 15

Question 16 (14 marks)

~~\$ 450 000~~

a) Michelle borrows \$50 000 to be repaid by regular monthly repayments of \$ P over a period of 25 years at 6% per annum reducible monthly. Interest is calculated and charged just before each repayment.

Let A_n be the amount owing after n -repayments.

(i) Show that the expression for the amount owing after two repayments is

1

$$A_2 = 450\,000(1.005)^2 - P(1.005) - P$$

$$A_1 = 450\,000 \left(1 + \frac{6 \div 12}{100}\right) - P$$

$$= 450\,000(1.005) - P$$

$$A_2 = A_1(1.005) - P$$

$$= 450\,000 \times 1.005^2 - P(1.005) - P$$

$\therefore$ shown

1 - correct solns.

(ii) Show that the amount owing after n -repayments is

2

$$A_n = 450\,000(1.005)^n - P \frac{(1.005)^n - 1}{0.005}$$

following pattern

$$\text{from (i)} \quad A_n = 450\,000(1.005)^n - P(1.005)^{n-1} - \dots - P$$

$$\therefore A_n = 450\,000(1.005)^n - P(1.005^{n-1} + \dots + 1)$$

G.P. $a=1$ $r=1.005$
 $n=n$

$$\therefore A_n = 450\,000(1.005)^n - P \frac{r^n - 1}{r - 1}$$

$$A_n = 450\,000(1.005)^n - P \frac{(1.005)^n - 1}{1.005 - 1}$$

$$\therefore A_n = 450\,000(1.005)^n - P \frac{(1.005)^n - 1}{0.005}$$

$\therefore$ shown

2 - correct soln.
1 - correctly applies pattern from (i)
1 - correctly applies the Sum of GP formula

Question 16 continued on the next page

(iii) Calculate the amount of each repayments P .

2

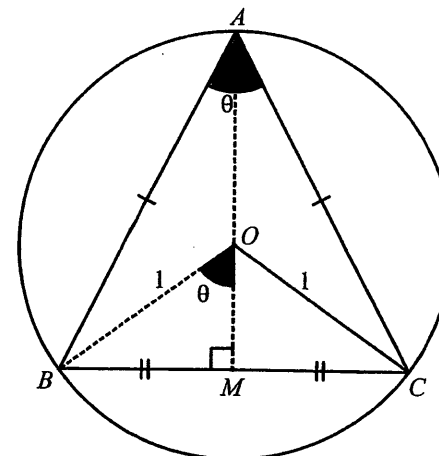
after 25 years = $25 \times 12 = 300 = n$ 2- correct soln.
 $\therefore A_{300} = 0 = 450000(1.005)^{300} - P \frac{1.005^{300} - 1}{0.005}$ 1- using part (i)
 correctly equates
 $A_{300} = 0$
 and attempts
 to solve it for P
 $P \frac{1.005^{300} - 1}{0.005} = 450000(1.005)^{300}$
 $P = \frac{450000(1.005)^{300}}{1.005^{300} - 1} \times 0.005$

$P = \$2899.3563...$

$\therefore P = \$2899.36$

Question 16 continued on the next page

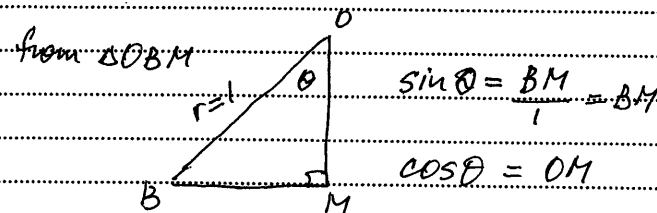
b) An isosceles triangle $\triangle ABC$ is inscribed within a unit circle centred at O , as shown in the diagram below. Let M be the midpoint of BC , $\angle BAC = \theta$ and $\angle BOM = \theta$.



(i) Show that the area of $\triangle ABC$ is $A = \sin\theta(1 + \cos\theta)$.

2

Area $\triangle ABC = \frac{1}{2} BC \times AM$ (AM $\perp$ BC) 2- correct soln.
 base height 1- finds BC or AM in terms of θ



$AM = AO + OM = 1 + \cos\theta$

$BC = 2 \times BM = 2 \times \sin\theta$

$\therefore \text{Area} = \frac{1}{2} \times BC \times AM = \frac{1}{2} \times 2 \sin\theta \times (1 + \cos\theta)$
 $\therefore A = \sin\theta(1 + \cos\theta) \therefore \text{shown}$

Question 16 continued on the next page

- (iii) Hence prove that the area of the isosceles triangle is maximum when it is equilateral.

3

3 - correct soln.
2 - finds stationary points correctly
1 - differentiates Area formula correctly & attempts to solve $A' = 0$
1 - determines the nature of st.pts and concludes for Δ to be equilateral

$$A = \sin \theta (1 + \cos \theta)$$

$$A' = \cos \theta (1 + \cos \theta) + \sin \theta (0 - \sin \theta)$$

$$\therefore A' = \cos \theta + \cos^2 \theta - \sin^2 \theta$$

$$A' = 0$$

$$0 = \cos \theta + \cos^2 \theta - \sin^2 \theta$$

$$0 = \cos \theta + \cos^2 \theta - (1 - \cos^2 \theta)$$

$$0 = 2\cos^2 \theta + \cos \theta - 1$$

$$\theta = (2\cos^2 \theta - 1)(\cos \theta + 1)$$

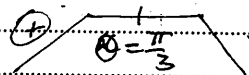
$$\cos \theta = \frac{1}{2} \quad \cos \theta = -1$$

$$\theta = \frac{\pi}{3}, \left(\frac{5\pi}{3}\right) \text{ - impossible in triangle } \theta \text{ - acute}$$

$$\theta = 180^\circ$$

$$\therefore \theta = \frac{\pi}{3} \text{ (the only solution)}$$

Nature	θ	1	$\frac{\pi}{3}$	1.1
A'		0.124	0	-0.135

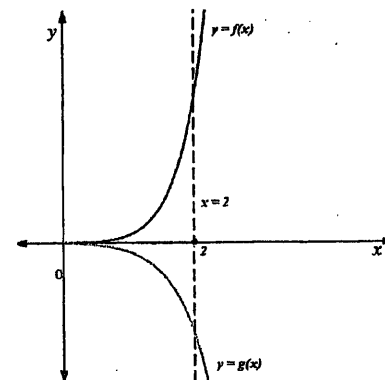


$\therefore$ at $\theta = \frac{\pi}{3}$ Area is Maximum

but if $\theta = \frac{\pi}{3}$ $\therefore \Delta ABC$ is equilateral/
and given ΔABC is isosceles

Question 16 continued on the next page

- c) The graph of $f(x) = x^2 e^{kx}$ and $g(x) = -\frac{2xe^{kx}}{k}$ and the line $x = 2$ is drawn below.
 $f(x) = g(x)$ at only one point, that is at $(0, 0)$.



Let A be the area of the region bounded by the curve $y = f(x)$, $y = g(x)$ and the line $x = 2$.

- (i) Write down a definite integral that gives the value of A .

1

1 - correct soln.

$$A = \int_0^2 f(x) - g(x) dx$$

Q.R.

$$\therefore A = \int_0^2 x^2 e^{kx} - \left(-\frac{2xe^{kx}}{k}\right) dx$$

- (ii) The function $f(x)$ from part (i) is given by $f(x) = x^2 e^{kx}$ where k is a positive constant. Show that $f'(x) = xe^{kx}(kx + 2)$

1 - correct soln.

$$f(x) = x^2 e^{kx} \quad k > 0$$

$$f'(x) = 2xe^{kx} + x^2 \cdot k e^{kx}$$

$$\therefore f'(x) = xe^{kx}(2 + kx)$$

$\therefore$ shown

Question 16 continued on the next page

(iii) Using the results of part (i) and (ii), or otherwise, find the value of k such that 2

$$A = \frac{16}{k}$$

from (i) $A = \int_0^2 x^2 e^{kx} - \frac{2xe^{kx}}{k} dx$

$$\therefore A = \int_0^2 x^2 e^{kx} + \frac{2xe^{kx}}{k} dx$$

$$= \frac{1}{k} \int_0^2 kx^2 e^{kx} + 2xe^{kx} dx$$

$$= \frac{1}{k} \int_0^2 x e^{kx} (kx + 2) dx$$

but $= f'(x)$

$$\therefore A = \frac{1}{k} \left[x^2 e^{kx} \right]_0^2$$

$f(x)$

$$\therefore A = \frac{16}{k}$$

2-correct solns.

1+ simplifies integrand

$$A = \frac{1}{k} \int_0^2 f'(x) dx$$

$$\therefore \frac{16}{k} = \frac{1}{k} \left[x^2 e^{kx} \right]_0^2$$

$$16 = [2^2 e^{2k} - 0]$$

$$16 = 4e^{2k}$$

$$4 = e^{2k}$$

$$\ln 4 = 2k$$

End of Exam

$$k = \frac{1}{2} \ln 4 \quad \text{or} \quad k = \ln \sqrt{4} = \ln 2$$